LoadCoherent: aggregation consistency and covariance uncertainty in synthetic district heat-load predictions

Reproducible computational study, 8 October 2026

Abstract

Aggregation constraints provide a useful check on district heat-load predictions, but satisfying them does not establish predictive accuracy. This study implements covariance-weighted reconciliation using Cholesky whitening and QR factorization. A synthetic four-building network permits analytical evaluation of prediction risk and independent verification by a constrained linear solve. Oracle weights reduce the sum of node mean-square errors by 41.19%, whereas a deliberately misspecified weight increases it by 8.25% despite coherence. Independent finite-calibration experiments show a tradeoff: fixed diagonal shrinkage limits small-sample variance, but retains estimation bias as sample size grows. The study separates numerical projection error, statistical estimation error and untested physical validity. It evaluates an established method rather than proposing a new forecasting algorithm.

1 Question and thermal assumptions

The research question is: when is an aggregation-consistent heat-load prediction less accurate than its unconstrained input? This is relevant when building, district and plant forecasts are generated separately. Topic selection arose from DTU Advanced Time Series Analysis lecture material on hierarchical reconciliation and heat-load forecasting. All network parameters, experiments, code and illustrations here are original synthetic constructions; no private course material or measured load data is reproduced.

Four buildings are partitioned into north (1,2) and south (3,4). Building demand is the quasi-steady heat loss

$$b_i(t) = U_i[T_{\text{in}} - T_{\text{out}}(t)], \quad T_{\text{out}}(t) = 4 + 5\sin(2\pi t/24), \quad T_{\text{in}} = 21. \tag{1}$$

Temperatures are in degrees Celsius, t in hours, and $U = (1.2, 0.8, 1.5, 1.0)$ in kW/K. Perfect demand tracking, negligible storage and lossless pipes are assumed. With node order plant, north, south, 1,2,3,4,

$$y = Sb, \qquad S = \begin{bmatrix} S_{\text{top}} \\ I_4 \end{bmatrix} \in \mathbb{R}^{7\times 4}, \qquad S_{\text{top}} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}. \tag{2}$$

The last four rows are the identity. Equation (1) is an algebraic thermal model: it neither resolves transient conduction nor provides a district heating design calculation. Its role is to supply a coherent, positive reference signal for evaluating a statistical projection.

2 Reconciliation and analytical risk

Let $\hat{y} = y + e$, with $\mathbb{E}e = \mu$ and $\text{Cov}(e) = V$. For any symmetric positive-definite weight W, define

$$\tilde{y} = S \arg\min_z (Sz - \hat{y})^T W^{-1} (Sz - \hat{y}) = P_W \hat{y}. \tag{3}$$

This is the weighted reconciliation framework of Wickramasuriya et al. [1]; a public exposition is given by Hyndman and Athanasopoulos [2]. For full-column-rank S, $P_W S = S$ and $P_W^2 = P_W$. Thus the exact signal is preserved, and errors transform as $P_W e$. Taking the second moment gives the per-node risk used here:

$$r_j = [P_W V P_W^T]_{jj} + [P_W \mu]_j^2, \qquad R = \sum_{j=1}^{7} r_j. \tag{4}$$

3 Implementation and reproducible experiments

3.1 Algorithm and verification

For $W = LL^T$, form $A = L^{-1}S = QR$ by Cholesky whitening and reduced QR. Solve $RG = Q^T L^{-1}$ with triangular solves and set $P_W = SG$. Application uses $S(G\hat{y})$, or its transposed batch equivalent. No explicit inverse or normal-equation matrix is constructed. The dense Python implementation uses NumPy and SciPy. Factoring weights costs $O(n^3)$; this implementation targets small hierarchies, and no speed advantage is claimed. It rejects nonfinite or complex data, incompatible shapes, relative weight eigenvalues at or below 10^{-12}, and relative whitened singular values at or below 10^{-12}. These cutoffs are safeguards rather than accuracy certificates.

An independent saddle-point computation solves

$$\begin{bmatrix} V & S \\ S^T & 0 \end{bmatrix} \begin{bmatrix} \lambda \\ z \end{bmatrix} = \begin{bmatrix} \hat{y} \\ 0 \end{bmatrix}, \qquad \tilde{y} = Sz, \tag{5}$$

for the oracle case. Tests additionally cover a scalar inverse-variance example, weighted orthogonality, scale invariance, batch application, projection identities, covariance estimation, invalid inputs and coherent bias. The installed-wheel tests and complete numerical-output reproduction run on Windows and Ubuntu in CI. The manuscript build reads numerical tables directly from the results JSON, records input/PDF hashes and enforces a five-page limit.

3.2 Specified error population

Independent Gaussian error vectors use $V = FF^T + \mathrm{diag}(d^2)$, where

$$F = \begin{bmatrix} 7 & 0 & 0 \\ 2 & 4 & 0 \\ 3 & 0 & 3 \\ 1 & 2 & 0 \\ 1.5 & -1 & 0 \\ 1 & 0 & 1.5 \\ 2 & 0 & -1 \end{bmatrix}, \qquad d = \begin{bmatrix} 3 & 2 & 2.5 & 1 & 1.2 & 1.1 & 1.5 \end{bmatrix}^T. \tag{6}$$

Factor coefficients and d are in kW. Shared factors represent correlated forecast errors, not simulated random heat sources. They are deliberately not coherent across aggregation levels. The positive diagonal term ensures V is positive definite. No forecasting model is trained: the experiment supplies its prediction errors directly, isolating reconciliation from model selection.

Five fixed references are evaluated: unreconciled base; bottom up using only the last four forecasts; OLS ($W = I$); oracle ($W = V$); and wrong weights $W = \mathrm{diag}(0.01, 1, 1, 100, 100, 100, 100)$. The last choice overtrusts the plant and is a deliberate stress test. For $\mu = 0$, equation (4) gives the exact population risk. The sum includes overlapping aggregate targets; it is a forecasting loss, not a conserved thermal quantity. Node-level results are also recorded.

3.3 Calibration and sampling protocol

NumPy seed 20261008 spawns separate test, calibration and display generators. For each sample size $N \in \{10, 20, 50, 200\}$, 80 independent calibration sets are drawn. The same set serves three estimators for a paired comparison:

$$\hat{V} = \frac{1}{N-1} \sum_{k=1}^{N} (e_k - \bar{e})(e_k - \bar{e})^T, \qquad W_\alpha = (1-\alpha)\hat{V} + \alpha\, \mathrm{diag}(\hat{V}), \tag{7}$$

with $\alpha = 0, 1/2, 1$. Half-shrinkage is fixed in advance, not estimated to minimize reported test risk. Each fitted P_W is evaluated analytically against the known V, so conditional risks have no test-sample Monte Carlo noise. Separately, 60,000 test errors validate the five fixed-method risks. For these zero-mean Gaussian errors, the analytic standard error of the mean total squared loss is $[2\,\mathrm{tr}\{(P_W V P_W^T)^2\}/60000]^{1/2}$. The illustrative load path uses 48 hourly points and independent error draws; it is not a temporally correlated forecast simulation.

4 Measured results

Table 1: Fixed weights: analytical risk and independent Monte Carlo check. Risk and sampled MSE have units kW^2; z is the discrepancy divided by its analytic Monte Carlo standard error.

Method	Exact R	R/R_{base}	Sampled MSE	z
Base	128.6500	1.000000	128.7995	0.304
Bottom up	82.2000	0.638943	81.8688	-0.859
OLS	104.7000	0.813836	104.7081	0.017
Oracle	75.6632	0.588132	75.2828	-1.014
Wrong weights	139.2623	1.082489	139.4650	0.329

Oracle weights reduce expected all-node error by 41.19%, but wrong weights increase it by 8.25%. Thus coherence alone does not imply improvement. The largest absolute Monte Carlo discrepancy is 1.014 standard errors. The QR/KKT forecast difference is 1.421×10^{-14} kW and the maximum entry of $P^2 - P$ is 4.441×10^{-16}. Across the illustrative path, the maximum absolute aggregation residual decreases from 11.8785 kW to 1.776×10^{-14} kW.

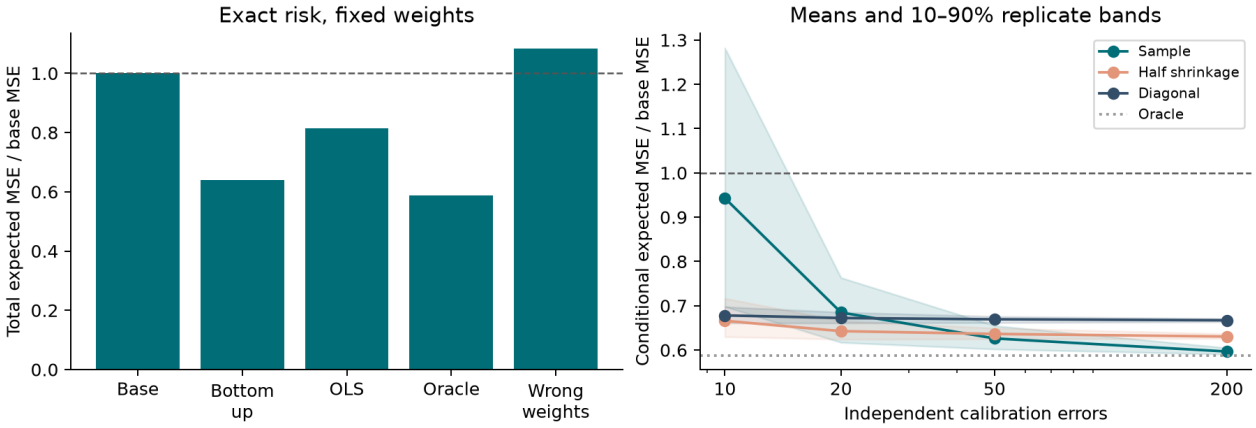


Figure 1: Exact risk of fixed methods (left) and calibration sensitivity (right). Lines show means over 80 calibration sets; shaded areas show their 10th–90th percentiles, not confidence intervals for the means.

Table 2: Conditional risk ratio across independent calibration sets. SE is the standard error of the replicate mean; the final column is the observed fraction of sets with risk above unreconciled base risk.

N	Estimator	Mean	SE	10th–90th percentile	Worse
10	Sample	0.9429	0.0386	0.6981–1.2816	25.00%
10	Half shrinkage	0.6661	0.0040	0.6300–0.7167	0.00%
10	Diagonal	0.6783	0.0022	0.6609–0.6976	0.00%
20	Sample	0.6848	0.0083	0.6172–0.7634	1.25%
20	Half shrinkage	0.6427	0.0017	0.6244–0.6661	0.00%
20	Diagonal	0.6724	0.0011	0.6610–0.6854	0.00%
50	Sample	0.6263	0.0027	0.6021–0.6551	0.00%
50	Half shrinkage	0.6365	0.0010	0.6250–0.6496	0.00%
50	Diagonal	0.6695	0.0007	0.6616–0.6760	0.00%
200	Sample	0.5968	0.0006	0.5906–0.6051	0.00%
200	Half shrinkage	0.6305	0.0005	0.6259–0.6368	0.00%
200	Diagonal	0.6673	0.0003	0.6635–0.6711	0.00%

At $N = 10$, the sample estimator exceeds base risk in 20 of 80 sets. Neither shrinkage estimator does so in these recorded sets, which does not prove a universal bound. At $N = 200$, the sample estimator has lower mean risk than fixed half-shrinkage. This reversal reflects covariance estimation variance at small N and persistent distortion from a fixed diagonal shrinkage target at large N. The study does not optimize the shrinkage fraction or extrapolate beyond the specified population and sizes.

5 Discussion, limitations and reproducibility

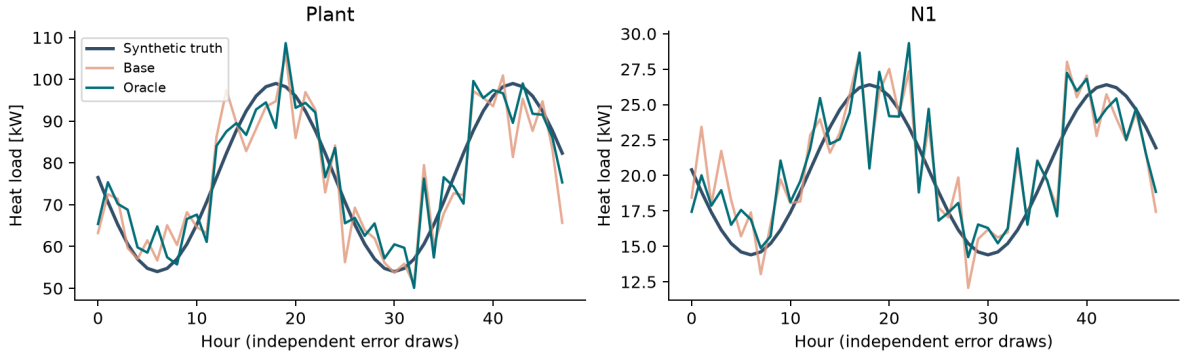


Figure 2: Illustrative synthetic demand at the plant and one building. Oracle reconciliation uses known error covariance. This single path visualizes the construction; the analytical risk study supports accuracy comparisons.

Bias is a separate problem. Any coherent bias $\mu = S\delta$ survives: $P_W \mu = \mu$. Setting every component of δ to 2 kW gives $\mu = (8, 4, 4, 2, 2, 2, 2)^T$ kW. The oracle risk then becomes 187.6632 kW^2, including a 112 kW^2 bias penalty. Centering calibration errors estimates covariance but does not subtract bias from subsequent forecasts. Linear reconciliation can also produce negative loads; a small counterexample is included in the test suite. Nonnegative constraints would require a different optimization problem.

Discrete error and continuum error. The KKT comparison and projection identities verify finite-dimensional numerical calculations to roundoff scale. Finite-calibration risk concerns statistical estimation, not floating-point error. Equation (1) is evaluated analytically without spatial or temporal discretization; there is no continuum mesh-error estimate or PDE convergence claim. The hourly display is a sampling choice, not a time integrator.

Physical validity and scope. No measurements validate these results. Pipe storage/losses, uncertain topology, controls, occupancy and time-correlated errors would change the model. Only one hierarchy and one fixed Gaussian covariance are studied. The oracle needs unknown population information. Replicate fractions are finite-experiment outcomes; half-shrinkage is not asserted to be optimal.

Reproduction. Install and execute from the repository root:

```
python -m pip install -e ".[dev]"
loadcoherent -output examples/results
python scripts/check_samples.py
python -m pytest -q
python paper/build.py -engine pdflatex
```

CSV files contain calibration risks and paths; JSON contains per-node risks, matrices and diagnostics. CI checks compilation and page count; hashes bind the paper to the numerical inputs. Every rendered page must also be inspected.

6 Conclusion

Aggregation checks and accuracy checks answer different questions. Weighted QR enforces the declared constraints, while covariance choice determines risk. Fixed shrinkage helps the smallest calibration set; sample covariance performs better at the largest tested size. Neither result validates a real network.

References

[1] S. L. Wickramasuriya, G. Athanasopoulos and R. J. Hyndman. Optimal forecast reconciliation for hierarchical and grouped time series through trace minimization. *Journal of the American Statistical Association*, 114(526):804–819, 2019. doi:10.1080/01621459.2018.1448825.

[2] R. J. Hyndman and G. Athanasopoulos. *Forecasting: Principles and Practice*, 3rd ed., OTexts, 2021. Section 11.3: Forecast reconciliation.